



MINING DEALER BEST PRACTICE

Final Drive Gear-Coupling Assembly Tool for Multiple HEX

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Final Drive Gear-Coupling Assembly Tool for Multiple HEX	1
1.0 Introduction	2
2.0 Best Practice Description	2
3.0 Implementation Steps	4
4.0 Benefits	4
5.0 Resources Required	4
6.0 Supporting Attachments / References	4
7.0 Related Best Practices	4
8.0 Acknowledgements	4

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1.0 Introduction

Component repair center (CRC) receives diverse customer's components daily and it is very important to check the correct position of the assemble parts aiming to prevent failures related to assembly mistaken.

Since Caterpillar has distinguished left and right side of Housing Motor (425-3520 RH / 425-3521 LH), it is required to have close attention to assembly process, once it is possible to assembly Gear Coupling (199-4719) in the mistaken position.

Image 1 shows Gear Coupling assembled in the mistaken position.

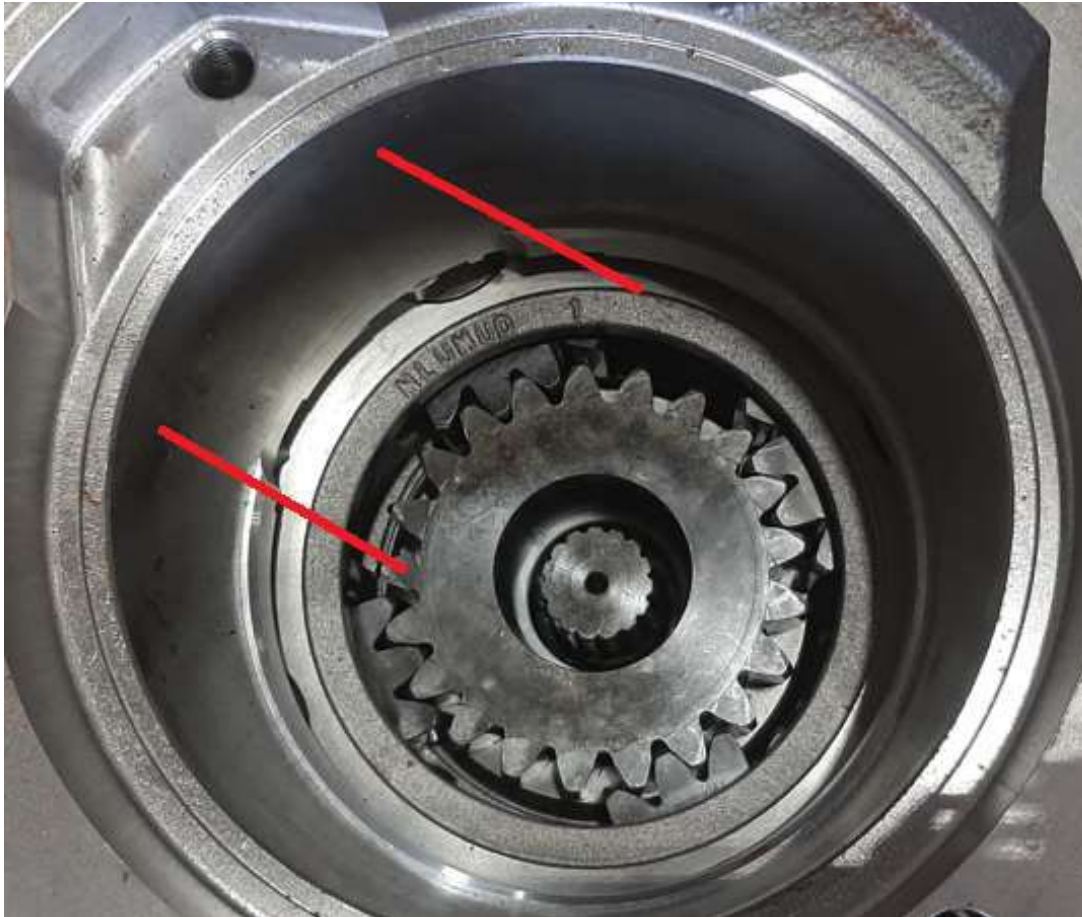


Image 1 – Gear-Coupling mistaken assembly.

Thus, CRC technicians developed a tool which avoids the mistaken assembly of the final drive gear-coupling 199-4719.

2.0 Best Practice Description

This Best Practice proposes to use an old Body AS 542-1323 of the Motor GP-Travel from a down equipment as a standard. In this case, it was defined to paint the structure in blue to differentiate it as a tool, as shown in image 2:

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Image 2 – Gauge assembly tool.

Then, technicians use it as a gauge in the bottom of the Housing Motor in both left and right sides, orienting the flange to the correct position and, consequently, providing a correct installation of the Gear Coupling (199-4719).



Image 3 – Example of a gauge tool use in the workshop.

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	12/9/2022	00	1.04-221209-1

3.0 Implementation Steps

1. Define an old Body AS of the Motor GP-Travel PN 542-1323 from a down equipment as a standard.
2. Choose a different color to paint the part due to set it as a tool.
3. Insert the tool in the bottom of the Housing Motor 425-3520 RH / 425-3521 LH as a gauge.
4. Put the Gear-coupling PN 199-4719 on the top of the Housing Motor matching the flange.
5. Check if the flange position of both parts are matching correctly.
6. Position the bolts in their position.
7. Torque the bolts using the specified measure.
8. Remove the tool.

4.0 Benefits

- Certify the correct assembly of the Gear Coupling (199-4719) in the Final Drive GP avoiding assembly mistakes.

5.0 Resources Required

- An old Body AS of the Motor GP-Travel PN 542-1323.
- Paint.

6.0 Supporting Attachments / References

N.A.

7.0 Related Best Practices

N.A.

8.0 Acknowledgements

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Final Drive Gear-Coupling Assembly Tool for Multiple HEX	DATE 12/9/2022	CHG NO 00	NUMBER 1.04-221209-1